

Concept: Eight supplementary demand/band calibration tasks

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# Eight supplementary demand/band calibration tasks: Construction

**Contents and concept route**



## Classify after putting the equation in standard form

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Classify after putting the equation in standard form

**Concept difficulty:** Medium concept

### **1** Classify after putting the equation in standard form

**Medium concept**

Section 1 of 8

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## Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

### 1 Collect the variable terms.

**Why valid:** Adding  $2x$  to both complete sides preserves equality.

**Result:**  $5x - 4 = 1$

### 2 Remove the constant.

**Why valid:** Adding 4 to both sides is reversible.

**Result:**  $5x = 5$

### 3 Select the category.

**Why valid:** The coefficient 5 is nonzero, so exactly one rational can satisfy the equation.

**Result:** One solution.

### 4 Challenge an all-solutions claim.

**Why valid:** One counterexample suffices to refute that every rational works.

**Result:**  $x=0$  gives  $-4$  and  $1$ , which differ.

Requested output: category of solution set

$$3x - 4 = 1 - 2x$$

No solution | One solution | Every rational

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### The key step

The first has every rational; the second has none. The reduced constant comparison matters when the coefficient is zero.

### Worked explanation

Over the rationals, classify  $3x - 4 = 1 - 2x$  as having no solution, one solution, or every rational as a solution. Use the criterion  $Ax = B$ :  $A$  nonzero gives one rational solution,  $A = B = 0$  gives every rational, and  $A = 0$  with  $B$  nonzero gives none. Do not report the value of  $x$ .

**Follow the operation, its justification and its result.**

1. **Collect the variable terms.**

Why valid: Adding  $2x$  to both complete sides preserves equality.

Result:  $5x - 4 = 1$

2. **Remove the constant.**

Why valid: Adding 4 to both sides is reversible.

Result:  $5x = 5$

3. **Select the category.**

Why valid: The coefficient 5 is nonzero, so exactly one rational can satisfy the equation.

Result: One solution.

4. **Challenge an all-solutions claim.**

Why valid: One counterexample suffices to refute that every rational works.

Result:  $x=0$  gives  $-4$  and  $1$ , which differ.

### Result

One solution.

### Check

At  $x=0$  the two sides differ, so the category is not every rational. The nonzero coefficient rules out none.

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

At  $x=0$  the two sides differ, so the category is not every rational. The nonzero coefficient rules out none.

### Now you do one

Classify  $4x+1=4x+1$  and  $4x+1=4x+2$ . Do not solve by dividing by the cancelled coefficient.

## Classify linked parameter exceptions

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Classify linked parameter exceptions

**Concept difficulty:** Medium concept

### 1 Classify linked parameter exceptions

Medium concept

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### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

#### 1 Expand E1 and cancel 2x.

**Why valid:** Distribution and subtracting the same term from both sides preserve the solution set.

**Result:** E1 becomes  $-6=k$ .

**2 Classify E1.**

**Why valid:** The reduced statement has no  $x$ , so it holds for all  $x$  or for none.

**Result:** Every rational if  $k = -6$ ; none otherwise.

**3 Factor the E2 constant.**

**Why valid:**  $4 - 2k = -2(k - 2)$ .

**Result:**  $(k - 2)x = -2(k - 2)$ .

**4 Separate  $k=2$  before division.**

**Why valid:** At  $k=2$  both sides vanish; division by  $k-2$  would lose this case.

**Result:** E2 has every rational if  $k=2$ .

**5 Classify the remaining  $k$ .**

**Why valid:** A nonzero rational coefficient has a unique rational quotient.

**Result:** E2 has one rational solution for  $k \neq 2$ .

**6 Cross-check the exceptional parameters.**

**Why valid:** Each category must be evaluated in the original pair.

**Result:** At  $k = -6$ : E1 every, E2 one. At  $k = 2$ : E1 none, E2 every.

$k$  is fixed;  $x$  ranges over rationals

$$E1: 2(x-3) = 2x+k$$

$$E2: (k-2)x = 4-2k$$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### The key step

At  $t=3$  every rational works; at  $t=4$  exactly one works. Split a zero divisor before cancelling.

### Worked explanation

Let  $k$  be rational and  $x$  rational. For each  $k$ , classify each of these equations as having none, one, or every rational solution: E1:  $2(x-3)=2x+k$ ; E2:  $(k-2)x=4-2k$ . Use the  $Ax=B$  criteria; a numerical solution is not requested.

**Follow the operation, its justification and its result.**

1. **Expand E1 and cancel  $2x$ .**

Why valid: Distribution and subtracting the same term from both sides preserve the solution set.

Result: E1 becomes  $-6=k$ .

2. **Classify E1.**

Why valid: The reduced statement has no  $x$ , so it holds for all  $x$  or for none.

Result: Every rational if  $k=-6$ ; none otherwise.

3. **Factor the E2 constant.**

Why valid:  $4-2k=-2(k-2)$ .

Result:  $(k-2)x=-2(k-2)$ .

4. **Separate  $k=2$  before division.**

Why valid: At  $k=2$  both sides vanish; division by  $k-2$  would lose this case.

Result: E2 has every rational if  $k=2$ .

5. **Classify the remaining  $k$ .**

Why valid: A nonzero rational coefficient has a unique rational quotient.

Result: E2 has one rational solution for  $k \neq 2$ .

6. **Cross-check the exceptional parameters.**

Why valid: Each category must be evaluated in the original pair.

Result: At  $k = -6$ : E1 every, E2 one. At  $k = 2$ : E1 none, E2 every.

**Result**

E1: every rational at  $k = -6$ , none otherwise. E2: every rational at  $k = 2$ , one otherwise.

**Check**

Substitute  $k = -6$  and  $k = 2$  in both originals. Neither equation has a hidden division step at its exceptional parameter.

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**1-2-3 QUICK CHECK**

**1 · CHECK**

Substitute  $k = -6$  and  $k = 2$  in both originals. Neither equation has a hidden division step at its exceptional parameter.

### Now you do one

Classify  $(t-3)y=2(t-3)$  at  $t=3$  and  $t=4$ , keeping the exceptional case.

## Classify with a coupled domain boundary

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Classify with a coupled domain boundary

**Concept difficulty:** Medium concept

### 1 Classify with a coupled domain boundary

Medium concept

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### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

**1 Exclude the parameter making the second equation undefined.**

**Why valid:** Its right denominator  $k-2$  must be nonzero.

**Result:**  $k=2$  is INVALID PARAMETER.

**2 Isolate the zero-factor case  $k=1$ .**

**Why valid:** The first equation then reads  $0=0$ , while the second remains defined.

**Result:** The second becomes  $(x+1)/(x-1)=-1$ , giving  $x=0$ ; it is rational and allowed.

**3 Reduce the first equation for  $k$  outside  $\{1,2\}$ .**

**Why valid:** Only in this branch is cancellation of  $k-1$  followed by division by  $k-2$  valid.

**Result:**  $x=1/(k-2)$ .

**4 Reduce the second equation with both denominator conditions retained.**

**Why valid:** Cross-multiplication is reversible when  $x \neq 1$  and  $k \neq 2$ .

**Result:**  $(k-2)(x+1)=k(x-1)$ , hence  $x=k-1$ .

**5 Require the two candidates to coincide.**

**Why valid:** Both original equations must hold for the same  $x$ .

**Result:**  $(k-1)(k-2)=1$ , hence  $k^2-3k+1=0$ .

**6 Use the rational-domain restriction.**

**Why valid:** The only compatibility parameters  $(3 \pm \sqrt{5})/2$  are irrational;  $x=k-1$  is then irrational and excluded.

**Result:** No rational solution for every  $k$  outside  $\{1,2\}$ .

**7 Verify the exceptional allowed branch in the originals.**

**Why valid:** A reduced-line result is insufficient when divisions were used in other branches.

**Result:**  $k=1, x=0$  gives  $0=0$  and  $1/(-1)=1/(-1)$ . ONE at  $k=1$ ; INVALID at  $k=2$ ; NONE otherwise; never EVERY.

$k$  real;  $x$  rational; both equations required

$$(k-1)(k-2)x = k-1$$

$$(x+1)/(x-1) = k/(k-2); \text{ denominators nonzero}$$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### The key step

At  $a=2$  the equation is defined but inconsistent (no solution); at  $a=3$  exactly one rational works; at  $a=2+\sqrt{2}$  the sole real candidate  $1/\sqrt{2}$  is irrational, so no rational works. Domain checks come after valid branch reduction.

### Worked explanation

Extension beyond routine linear equations.  $k$  is real,  $x$  must be rational. Classify the simultaneous system by  $k$  as INVALID PARAMETER, NO RATIONAL SOLUTION, ONE RATIONAL SOLUTION, or EVERY RATIONAL SOLUTION:  $(k-1)(k-2)x=k-1$  and  $(x+1)/(x-1)=k/(k-2)$ . An equation with a zero denominator is invalid; a solution must satisfy both original equations and  $x \neq 1$ . Report categories and the parameter cases, not a general numerical solution method.

### Follow the operation, its justification and its result.

1. **Exclude the parameter making the second equation undefined.**

Why valid: Its right denominator  $k-2$  must be nonzero.

Result:  $k=2$  is INVALID PARAMETER.

2. **Isolate the zero-factor case  $k=1$ .**

Why valid: The first equation then reads  $0=0$ , while the second remains defined.

Result: The second becomes  $(x+1)/(x-1)=-1$ , giving  $x=0$ ; it is rational and allowed.

3. **Reduce the first equation for  $k$  outside  $\{1,2\}$ .**

Why valid: Only in this branch is cancellation of  $k-1$  followed by division by  $k-2$  valid.

Result:  $x=1/(k-2)$ .

4. **Reduce the second equation with both denominator conditions retained.**

Why valid: Cross-multiplication is reversible when  $x \neq 1$  and  $k \neq 2$ .

Result:  $(k-2)(x+1)=k(x-1)$ , hence  $x=k-1$ .

5. **Require the two candidates to coincide.**

Why valid: Both original equations must hold for the same  $x$ .

Result:  $(k-1)(k-2)=1$ , hence  $k^2-3k+1=0$ .

6. **Use the rational-domain restriction.**

Why valid: The only compatibility parameters  $(3 \pm \sqrt{5})/2$  are irrational;  $x=k-1$  is then irrational and excluded.

Result: No rational solution for every  $k$  outside  $\{1,2\}$ .

## 7. Verify the exceptional allowed branch in the originals.

Why valid: A reduced-line result is insufficient when divisions were used in other branches.

Result:  $k=1, x=0$  gives  $0=0$  and  $1/(-1)=1/(-1)$ . ONE at  $k=1$ ; INVALID at  $k=2$ ; NONE otherwise; never EVERY.

### Result

ONE RATIONAL SOLUTION at  $k=1$ ; INVALID PARAMETER at  $k=2$ ; NO RATIONAL SOLUTION for every other real  $k$ ; never EVERY.

### Check

At  $k=1$ ,  $x=0$  satisfies both originals. At  $k=2$  a denominator is zero. At the two real compatibility roots,  $x$  is irrational, so a real solution does not meet the rational-domain request.

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

At  $k=1$ ,  $x=0$  satisfies both originals. At  $k=2$  a denominator is zero. At the two real compatibility roots,  $x$  is irrational, so a real solution does not meet the rational-domain request.

### Now you do one

Let  $y$  be rational and  $a$  real. Classify  $(a-2)y=1$  at  $a=2$ ,  $a=3$  and  $a=2+\sqrt{2}$ . Keep undefined operations separate from empty solution sets.

## Execute an inverse chain across a table and a process graph

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Execute an inverse chain across a table and a process graph

**Concept difficulty:** Medium concept

### 1 Execute an inverse chain across a table and a process graph

Medium concept

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### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

#### 1 Convert the exact observation.

**Why valid:** Ten millimetres are one centimetre.

**Result:**  $C=22/10=11/5$  centimetres.

**2 Undo  $C=(B+2)/5$ .**

**Why valid:** Multiplication by nonzero 5 and subtraction of 2 reverse this stage.

**Result:**  $B=5C-2=9$ .

**3 Undo  $B=3A+4$ .**

**Why valid:** Subtract 4, then divide by nonzero 3.

**Result:**  $A=(9-4)/3=5/3$ .

**4 Undo  $A=(X-1)/2$ .**

**Why valid:** Multiply by nonzero 2, then add 1.

**Result:**  $X=2(5/3)+1=13/3$  centimetres.

**5 Run the entire forward chain.**

**Why valid:** A correct recovered input must reproduce the original observation after every named operation.

**Result:**  $13/3 \rightarrow 5/3 \rightarrow 9 \rightarrow 11/5$  centimetres = 22 millimetres.

Observation table:  $C = 22$  mm; program uses cm

$X \rightarrow$  subtract 1, divide 2  $\rightarrow A$

$A \rightarrow$  multiply 3, add 4  $\rightarrow B \rightarrow$  add 2, divide 5  $\rightarrow C$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### The key step

These are the same observation. Both give  $B=8$ ,  $A=4/3$  and  $X=11/3$  cm. Different results reveal a unit mapping error.

### Worked explanation

A measuring program uses numerical readings in centimetres:  $X \rightarrow A=(X-1)/2 \rightarrow B=3A+4 \rightarrow C=(B+2)/5$ . The observation table gives  $C=22$  millimetres. The inverse procedure is prescribed: convert the observation to centimetres, undo the  $C$  operation, then  $B$ , then  $A$ . Execute it to recover  $X$  and verify every forward stage. Treat the displayed readings as exact.

### Follow the operation, its justification and its result.

1. **Convert the exact observation.**

Why valid: Ten millimetres are one centimetre.

Result:  $C=22/10=11/5$  centimetres.

2. **Undo  $C=(B+2)/5$ .**

Why valid: Multiplication by nonzero 5 and subtraction of 2 reverse this stage.

Result:  $B=5C-2=9$ .

3. **Undo  $B=3A+4$ .**

Why valid: Subtract 4, then divide by nonzero 3.

Result:  $A=(9-4)/3=5/3$ .

4. **Undo  $A=(X-1)/2$ .**

Why valid: Multiply by nonzero 2, then add 1.

Result:  $X=2(5/3)+1=13/3$  centimetres.

5. **Run the entire forward chain.**

Why valid: A correct recovered input must reproduce the original observation after every named operation.

Result:  $13/3 \rightarrow 5/3 \rightarrow 9 \rightarrow 11/5$  centimetres = 22 millimetres.

**Result**

$X=13/3$  centimetres.

**Check**

The forward chain gives  $A=5/3$ ,  $B=9$ ,  $C=11/5$  cm =22 mm, so neither a unit mistake nor a reversed-step mistake survives the check.

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**1-2-3 QUICK CHECK**

**1 · CHECK**

The forward chain gives  $A=5/3$ ,  $B=9$ ,  $C=11/5$  cm =22 mm, so neither a unit mistake nor a reversed-step mistake survives the check.

### Now you do one

Run the same inverse program with  $C=20$  millimetres, and separately with  $C=2$  centimetres. Should  $X$  differ?

## Choose the governing equation from its conditions

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Choose the governing equation from its conditions

**Concept difficulty:** Medium concept

### 1 Choose the governing equation from its conditions

Medium concept

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Next →

### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

**1 Read the coefficient condition.**

**Why valid:** The problem explicitly excludes  $A=0$ .

**Result:**  $A$  is nonzero.

**2 Select R1.**

**Why valid:** Division by the fixed nonzero rational  $A$  is defined and reversible.

**Result:** Use  $x=B/A$ .

**3 Check the alternative boundary.**

**Why valid:** The all-solutions rule needs  $A=B=0$  and therefore conflicts with the given  $A \neq 0$ .

**Result:** R2 does not apply.

Given:  $Ax = B$ ;  $A \neq 0$

Candidate R1: divide by A

Candidate R2: zero-coefficient identity

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### **The key step**

The first admits every rational, the second none. A rule is selected by both coefficients, not the word linear.

### Worked explanation

A linear calibration is stated as  $Ax=B$  with rational  $A$  and  $B$ , and  $A$  is known to be nonzero. Choose the governing rule for recovering  $x$ : R1,  $x=B/A$ ; R2, infer every rational is a solution from  $0=0$ . Only choose the applicable rule; do not compute a value.

**Follow the operation, its justification and its result.**

1. **Read the coefficient condition.**

Why valid: The problem explicitly excludes  $A=0$ .

Result:  $A$  is nonzero.

2. **Select R1.**

Why valid: Division by the fixed nonzero rational  $A$  is defined and reversible.

Result: Use  $x=B/A$ .

3. **Check the alternative boundary.**

Why valid: The all-solutions rule needs  $A=B=0$  and therefore conflicts with the given  $A \neq 0$ .

Result: R2 does not apply.

## Result

R1,  $x=B/A$ .

## Check

The boundary  $A=B=0$  would admit R2, showing why the stated nonzero condition determines this choice.

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### 1-2-3 QUICK CHECK

#### 1 · CHECK

The boundary  $A=B=0$  would admit R2, showing why the stated nonzero condition determines this choice.

## Now you do one

Choose the rule when  $A=0, B=0$ , and when  $A=0, B=1$ .

**Translate an ordered pair without changing meaning**

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Translate an ordered pair without changing meaning

**Concept difficulty:** Medium concept

## 1 Translate an ordered pair without changing meaning

Medium concept

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### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

#### 1 Map the table headings to coordinate slots.

**Why valid:** The convention gives the first slot to  $x$  and the second to  $y$ .

**Result:** Slots are  $(x,y)$ .

#### 2 Insert the two labelled values.

**Why valid:** The row assigns  $x=2$  and  $y=3$ .

**Result:**  $(2,3)$ .

#### 3 Check a swapped pair.

**Why valid:** Its first coordinate would claim  $x=3$ , contrary to the row.

**Result:** (3,2) represents a different point.

Table heading: x | y

Row: 2 | 3

Coordinate template: (horizontal, vertical)

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### The key step

They give (4,1) and (1,4), distinct points. Coordinate roles are part of the meaning.

### Worked explanation

The ordered-pair table has columns  $x$  and  $y$  and one row  $x=2$ ,  $y=3$ . Write that row as a point in Cartesian ordered-pair notation. The coordinate convention is horizontal  $x$  first and vertical  $y$  second; no equation solving is requested.

**Follow the operation, its justification and its result.**

1. **Map the table headings to coordinate slots.**

Why valid: The convention gives the first slot to  $x$  and the second to  $y$ .

Result: Slots are  $(x,y)$ .

2. **Insert the two labelled values.**

Why valid: The row assigns  $x=2$  and  $y=3$ .

Result:  $(2,3)$ .

3. **Check a swapped pair.**

Why valid: Its first coordinate would claim  $x=3$ , contrary to the row.

Result:  $(3,2)$  represents a different point.

### Result

$(2,3)$ .

## Check

Reading the pair back gives  $x=2, y=3$ , preserving both table headings.

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## 1-2-3 QUICK CHECK

### 1 • CHECK

Reading the pair back gives  $x=2, y=3$ , preserving both table headings.

## Now you do one

Translate rows  $x=4, y=1$  and  $x=1, y=4$ . Are their points identical?

## Order two dependent inverse steps

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Order two dependent inverse steps

**Concept difficulty:** Medium concept

## 1 Order two dependent inverse steps

Medium concept

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### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

#### 1 Read the forward dependency.

**Why valid:** The doubled quantity already includes the added 3.

**Result:** Start → add 3 → double → output.

#### 2 Reverse the outer operation first.

**Why valid:** Halving recovers the quantity before doubling, which still contains the added 3.

**Result:** Halve first, then subtract 3.

#### 3 Check the competing order on a changed start.

**Why valid:** An independent forward/reverse check tests the dependency rather than memorising a rule.

**Result:** Start 2 gives output 10; halve then subtract gives 2, subtract then halve gives 3.5.

Start → add 3 → double → 14

Inverse components: halve | subtract 3

Required: dependency-respecting order

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

### **The key step**

The first needs subtract 3 then halve; the second halve then subtract 3. The component list is unchanged, but the dependency changes.

## Worked explanation

A starting number is first increased by 3 and then doubled, producing 14. You know the inverse components: subtract 3, and halve. Coordinate them into the correct reverse order. State which component must come first and why; do not calculate the starting number.

**Follow the operation, its justification and its result.**

1. **Read the forward dependency.**

Why valid: The doubled quantity already includes the added 3.

Result: Start  $\rightarrow$  add 3  $\rightarrow$  double  $\rightarrow$  output.

2. **Reverse the outer operation first.**

Why valid: Halving recovers the quantity before doubling, which still contains the added 3.

Result: Halve first, then subtract 3.

3. **Check the competing order on a changed start.**

Why valid: An independent forward/reverse check tests the dependency rather than memorising a rule.

Result: Start 2 gives output 10; halve then subtract gives 2, subtract then halve gives 3.5.

## Result

Halve first, then subtract 3; undo the last operation before its input operation.

## Check

For a changed start 2, the forward output is 10; the selected inverse returns 2 while the competing order returns 3.5.

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#### 1-2-3 QUICK CHECK

##### 1 • CHECK

For a changed start 2, the forward output is 10; the selected inverse returns 2 while the competing order returns 3.5.

## Now you do one

For forward double then add 3, which inverse comes first? Compare with forward add 3 then double.

## Give the warrant for a reversible division

**Subject:** Mathematics

**Topic:** Algebraic decisions and domain boundaries

**Concept:** Give the warrant for a reversible division

**Concept difficulty:** Medium concept

### 1 Give the warrant for a reversible division

Medium concept

Section 8 of 8

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#### Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

#### 1 State the divisor property.

**Why valid:** The given divisor 2 is nonzero.

**Result:** Division by 2 is defined.

#### 2 Connect the operation to its inverse.

**Why valid:** Multiplying both sides back by 2 restores the same equality.

**Result:** The step is reversible and preserves the solution set, establishing  $x=4$ .

**3 Challenge the zero-divisor boundary.**

**Why valid:** Division by zero is undefined, so it has no reversible operation of this kind.

**Result:** The warrant cannot license division by zero.

Premise:  $2x = 8$

Claim:  $x = 4$

Requested: a warrant for the division step

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

## The key step

It becomes  $0=0$  and admits every real, so the non-reversible step enlarged the solution set. Identical operations need the relevant reversibility condition.

### Worked explanation

Over the reals, an author concludes  $x=4$  from  $2x=8$  by dividing both sides by 2. Supply one warrant explaining why the conclusion follows and why the same justification would fail for division by zero. No further equation solving is requested.

**Follow the operation, its justification and its result.**

1. **State the divisor property.**

Why valid: The given divisor 2 is nonzero.

Result: Division by 2 is defined.

2. **Connect the operation to its inverse.**

Why valid: Multiplying both sides back by 2 restores the same equality.

Result: The step is reversible and preserves the solution set, establishing  $x=4$ .

3. **Challenge the zero-divisor boundary.**

Why valid: Division by zero is undefined, so it has no reversible operation of this kind.

Result: The warrant cannot license division by zero.

### Result

Because 2 is nonzero, dividing both sides by 2 is defined and reversible, preserving equality and the solution set. Division by zero is undefined.

### Check

Multiplication by 2 takes  $x=4$  back to  $2x=8$ . The zero-divisor case lacks this reversible division.

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Multiplication by 2 takes  $x=4$  back to  $2x=8$ . The zero-divisor case lacks this reversible division.

**Now you do one**

From  $x=1$  multiply both sides by zero. Does the resulting equation have exactly the original solution?

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